DYNAMICAL TITS ALTERNATIVE FOR GROUPS OF ALMOST AUTOMORPHISMS OF TREES

MARTÍN GILABERT VIO

Abstract. We prove a dynamical variant of the Tits alternative for the group of almost automorphisms of a locally finite tree $\mathcal{T}$: a group of almost automorphisms of $\mathcal{T}$ either contains a nonabelian free group playing ping-pong on the boundary $\partial\mathcal{T}$, or the action of the group on $\partial\mathcal{T}$ preserves a probability measure. This generalises to all groups of tree almost automorphisms a result of S. Hurtado and E. Militon for Thompson's group V, with a hopefully simpler proof.

1. Context and contributions

The Tits alternative is a celebrated theorem by J. Tits [Tit72] that shows a sharp dichotomy for linear groups over a field of characteristic zero: either they are virtually solvable or they contain a non-abelian free group. A group G is said to satisfy the Tits alternative if for every subgroup H of G, H is virtually solvable or contains a non-abelian free group. This group property has been established for a great deal of countable groups (see the references in [dlH00, Section II B, Complement 42]), usually by applying the Klein ping-pong lemma to exhibit free subgroups.

There are also many countable groups known to fail this alternative, as do many groups of homeomorphisms of compact spaces. For instance, the group $\mathrm{Homeo}(S^1)$ of homeomorphisms of the circle and the group of automorphisms $\mathrm{Aut}(\mathcal{T})$ of a regular tree $\mathcal{T}$ of degree ≥ 3: the former contains Thompson's group F of piecewise dyadically affine homeomorphisms of $[0,1]$, the latter contains the first Grigorchuk group, and these subgroups are not virtually solvable and do not contain free groups (see [CFP96] and [Gri80], respectively). Nevertheless, these two examples satisfy a dynamical variant of this condition which we formulate as follows.

Definition 1.1. *Let X be a compact topological space and G a group of homeomorphisms of X. We say that the action of G on X* satisfies the dynamical Tits alternative *if for every subgroup H of G one of the following holds.*

- *The action of H preserves a regular probability measure on X.*
- *There exists a* ping-pong pair *for the action of H, that is, there exist $g, h \in H$ and disjoint open sets $U_1, U_2, V_1, V_2 \subseteq X$ such that*

$$g(X \setminus U_1) \subseteq V_1 \quad \text{and} \quad h(X \setminus U_2) \subseteq V_2. \tag{1.1}$$

Date: April 30, 2026.

2020 *Mathematics Subject Classification.* Primary 20F99; Secondary 20F65.

Key words and phrases. Neretin group, Tits alternative, Thompson groups.

This dynamical alternative is a property of a group action, not merely of a group. Nonetheless, if $g, h \in H$ belong to a ping-pong pair, the ping-pong lemma shows that g, h generate a non-abelian free group. Moreover, the conditions in Definition 1.1 exclude each other and it suffices to verify them on finitely generated subgroups of G, see the beginning of the proof of Theorem A.

Remark. Previous work [MM23, HM19] involving this notion define the dynamical Tits alternative as a weaker condition, where every subgroup H is required to preserve a probability measure or to contain a non-abelian free group. We prefer our definition since this weaker notion is not a dichotomy, and moreover all known proofs of the alternative yield the stronger condition. For instance, whenever $G \curvearrowright X$ satisfies Definition 1.1, a subgroup H of G preserves a probability measure on X if and only if every pair of elements of H preserve a common probability measure on X.

Examples. A first family of examples comes from one-dimensional dynamics: the action on S^1 of the group of homeomorphisms of S^1 satisfies the dynamical Tits alternative by a theorem of G. Margulis [Mar00]. A related example is the Higman-Thompson group V acting on the triadic Cantor set, which also satisfies the alternative by work of Hurtado-Militon [HM19, Theorem 1.3]. A generalization of both statements is given in [MM23, Theorem 1.3], where it is shown that for any compact $K \subseteq \mathbb{R}$, the defining action of the group of locally monotone homeomorphisms of K satisfies the alternative. It is notable that the proof in [MM23] finds sufficiently proximal elements on a group G that does not preserve a measure on K by running a random walk on G, whereas the arguments in [Mar00, HM19] are "deterministic". Groups acting by homeomorphisms on dendrites also satisfy the alternative by work of Duchesne-Monod [DM18, Theorem 1.6].

A second family of examples consists of groups acting on the boundary of Gromov-hyperbolic spaces: a first elementary instance of this class is the action of automorphism group of a locally finite tree $\mathcal{T}$ on its boundary $\partial\mathcal{T}$, as follows easily from the well-known dynamical classification of subgroups of $\mathrm{Aut}(\mathcal{T})$, see [Tit70]. More generally, if (M, d) is a Gromov-hyperbolic and proper metric space such that $\mathrm{Isom}(M, d)$ acts cocompactly on M, then the action of $\mathrm{Isom}(M, d)$ on the Gromov boundary ∂M satisfies the dynamical Tits alternative (see [CCMT15], and also [AS22, Theorem 1.10] for a probabilistic version).

This note is concerned with almost automorphism groups of locally finite trees, which is a large family of locally compact and totally disconnected groups that arise as follows: let $\mathcal{T}$ be a locally finite rooted tree and $\partial\mathcal{T}$ be its space of ends. The group of rooted tree automorphisms $\mathrm{Aut}_{\mathrm{r}}(\mathcal{T})$ acts on $\partial\mathcal{T}$ preserving the so-called visual metric. The group $\mathrm{AAut}(\mathcal{T})$ of almost automorphisms of $\partial\mathcal{T}$ consists of all homeomorphisms of $\partial\mathcal{T}$ that are local homothecies for this metric, that is, that locally rescale it. The natural group topology on $\mathrm{AAut}(\mathcal{T})$ is not the compact-open topology, but the unique group topology such that $\mathrm{Aut}(\mathcal{T})$ is a compact open subgroup of $\mathrm{AAut}(\mathcal{T})$. See Section 2 for more precise statements.

If $d \geq 2$, $k \geq 1$ and $\mathcal{T}_{d,k}$ is the rooted tree where the root has k children and all other vertices have d children, then $\mathrm{AAut}(\mathcal{T}_{d,k})$ is known as a Neretin group. These groups are known to be

simple [Kap99], compactly presented [LB17] and to contain the Higman-Thompson group $V_{d,k}$ as a dense subgroup. They are the first examples of compactly generated simple groups without lattices [BCGM12] and moreover admit no invariant random subgroups by a result of T. Zheng [Zhe19].

We show that the action of $\mathrm{AAut}(\mathcal{T})$ on $\partial\mathcal{T}$ satisfies the dynamical Tits alternative for any locally finite rooted tree $\mathcal{T}$.

Theorem A. *Let $\mathcal{T}$ be a locally finite rooted tree. The action of $\mathrm{AAut}(\mathcal{T})$ on the boundary $\partial\mathcal{T}$ satisfies the dynamical Tits alternative.*

Some interesting groups to which Theorem A applies are the groups V_G considered by V. Nekrashevych in [Nek18], where $G \subseteq \mathrm{Aut_r}(\mathcal{T}_{2,2})$ is a self-similar group. Here V_G is the subgroup of $\mathrm{AAut}(\mathcal{T}_{2,2})$ generated by G and Higman-Thompson's V.

Fix a linear order on $\partial\mathcal{T}$ that is compatible with the tree structure. We denote by $V_{\mathcal{T}}$ the group of elements of $\mathrm{AAut}(\mathcal{T})$ acting in a locally order-preserving manner. For instance, $V_{\mathcal{T}_{d,k}}$ is the Higman-Thompson group associated to $\mathcal{T}_{d,k}$, and all topological full groups of irreducible infinite one-sided shifts of finite type are naturally subgroups of some $V_{\mathcal{T}}$, see [Led20, Theorem 3.8]. We call $V_{\mathcal{T}}$ the *Higman-Thompson group associated to $\mathcal{T}$*, although this name is not standard. The proof of Theorem B below gives a shorter and hopefully more conceptual approach to [HM19, Theorem 1.3] when specialized to the Higman-Thompson group V.

Theorem B. *Let H be a finitely generated subgroup of $V_{\mathcal{T}}$. Then the action of H on $\partial\mathcal{T}$ has a finite orbit or admits a ping-pong pair.*

We emphasize that Theorem B is not new, since the proof of [HM19, Theorem 1.3] shows that V verifies the (slightly stronger) conclusion of Theorem B and general arguments allow to extend the result from V to any group $V_{\mathcal{T}}$.

Two important ingredients in both proofs are a characterization of relatively compact subgroups of $\mathrm{AAut}(\mathcal{T})$ by Le Boudec-Wesolek [LBW19] and a description of the dynamics of individual elements of $\mathrm{AAut}(\mathcal{T})$ by Goffer-Lederle [GL21] (building on work of O. Salazar-Díaz [SD10]).

Acknowledgements. The author wishes to thank Gil Goffer, Adrien Le Boudec, Waltraud Lederle, Nicolás Matte Bon, Mikaël de la Salle and the anonymous referee for useful comments on a previous draft of this preprint. The author also thanks his advisor Nicolás Matte Bon for advice and encouragement, and Alejandra Garrido for pointing out [GL21].

2. Preliminaries

We give some background on $\mathrm{AAut}(\mathcal{T})$, describe the dynamics of individual elements of $\mathrm{AAut}(\mathcal{T})$, recall the definition of the Vietoris topology on closed subsets of a space and fix some notation. For more details on this material, see [GL21, LBW19, GL18].

Notation. Given a metric space (X, d), $A \subseteq X$ and $\varepsilon > 0$, we denote $A^\varepsilon = \{x \in X : d(x, A) \leq \varepsilon\}$. When $\mathcal{T}$ is a rooted tree, we write $\mathrm{Aut_r}(\mathcal{T})$ for the group of tree automorphisms of $\mathcal{T}$ that fix the root.

Almost automorphism groups of trees. Let $\mathcal{T}$ be a locally finite rooted tree with no leaves. We denote its root by v_0, and assume that all edges are directed away from v_0. A *caret* is a subtree of $\mathcal{T}$ consisting of a vertex, its children, and the edges between them. A subtree is *complete* if it is a union of carets, and when $\mathcal{T}_1, \mathcal{T}_2$ are rooted complete subtrees of $\mathcal{T}$ we denote by $\mathcal{T}_2 \setminus \mathcal{T}_1$ the union of all carets in $\mathcal{T}_2$ that are not included in $\mathcal{T}_1$. The set of leaves of a tree $\mathcal{T}$ is denoted by $\mathcal{LT}$.

Let $\partial\mathcal{T}$ be the *space of ends* of $\mathcal{T}$, that is, the set of (one-sided) infinite directed paths starting at v_0. We equip $\partial\mathcal{T}$ with the topology induced by the *visual metric* defined as $d(\xi, \xi') = 2^{-N(\xi,\xi')}$ for $\xi, \xi' \in \partial\mathcal{T}$ where $N(\xi, \xi') \in \mathbb{N}$ is the smallest integer n such that $\xi_n \neq \xi'_n$. The space $(\partial\mathcal{T}, d)$ is totally disconnected and compact, and its topology has a basis of clopen balls given by the sets $\partial\mathcal{T}_v = \{\xi \in \partial\mathcal{T} : v \text{ is in } \xi\}$ where $v \in V(\mathcal{T})$.

An *almost automorphism* of $\mathcal{T}$ is a homeomorphism g of $\partial\mathcal{T}$ such that there exists a partition of $\partial\mathcal{T}$ into clopen balls $D_1, \ldots, D_n$ and positive numbers $\lambda_1, \ldots, \lambda_n$ such that $d(g(y), g(z)) = \lambda_j d(y, z)$ for all $y, z \in D_j$. Such a partition is said to be *admissible* for g. Another way of viewing an almost automorphism is the following: take $\mathcal{T}_1, \mathcal{T}_2$ finite subtrees of $\mathcal{T}$ with root v_0, so $\mathcal{T} \setminus \mathcal{T}_1$, $\mathcal{T} \setminus \mathcal{T}_2$ are naturally rooted forests. Then any isomorphism $\overline{g} \colon \mathcal{T} \setminus \mathcal{T}_1 \to \mathcal{T} \setminus \mathcal{T}_2$ of rooted forests determines a $g \in \mathrm{AAut}(\mathcal{T})$, and conversely any almost automorphism arises in this manner, although not uniquely so.

Call an almost automorphism $g \in \mathrm{AAut}(\mathcal{T})$ *elliptic* if there exists a partition $\mathcal{P}$ of $\partial\mathcal{T}$ into clopen balls that is admissible for g and g-invariant, that is, such that $g\mathcal{P} = \mathcal{P}$. The following theorem is stated for Neretin groups in [LBW19], but the proof for any locally finite tree $\mathcal{T}$ is the same word by word.

Proposition 2.1 ([LBW19, Corollary 3.6])**.** *Let H be a finitely generated subgroup of* $\mathrm{AAut}(\mathcal{T})$*. The following are equivalent.*

- *H is relatively compact in* $\mathrm{AAut}(\mathcal{T})$*.*
- *Every element of H is elliptic.*
- *There is a partition $\mathcal{P}$ of $\partial\mathcal{T}$ into clopen balls such that $\mathcal{P}$ is admissible for every $h \in H$.*

Moreover, if these conditions hold and $\mathcal{Q}$ is any partition of $\partial\mathcal{T}$ into clopen balls such that $\mathcal{Q}$ is admissible for every element of some generating set of H, then $\mathcal{P}$ can be chosen to be finer than $\mathcal{Q}$.

Fix a family of total linear orders $\{<_v : v \in V(\mathcal{T})\}$ indexed by the vertices of $\mathcal{T}$, where $<_v$ is an order on the children of $v \in V(\mathcal{T})$. This family induces a linear order $<$ on $\partial\mathcal{T}$ by declaring $\xi < \xi'$ if $\xi, \xi' \in \partial\mathcal{T}$ and $\xi_n <_{\xi_{n-1}} \xi'_n$, where $n = N(\xi, \xi') \in \mathbb{N}$. We call $g \in \mathrm{AAut}(\mathcal{T})$ a *Higman-Thompson element* if there exists an admissible partition $\mathcal{P}$ for g such that $g\big|_D$ is order-preserving for all $D \in \mathcal{P}$. The subset of Higman-Thompson elements of $\mathrm{AAut}(\mathcal{T})$ is a group that we denote by $V_{\mathcal{T}}$.

We omit $\{<_v : v \in V(\mathcal{T})\}$ from the definition of $V_{\mathcal{T}}$ to keep the notation uncluttered, but for non-regular trees $\mathcal{T}$ the group $V_{\mathcal{T}}$ may depend on the choice of these orders.

Since the elliptic elements of $V_{\mathcal{T}}$ are exactly the elements of finite order in $V_{\mathcal{T}}$, as in [LBW19] Proposition 2.1 yields the following result.

Corollary 2.2 ([LBW19, Corollary 3.7])**.** *Any finitely generated subgroup of $V_{\mathcal{T}}$ composed entirely of elliptic elements is finite.*

Dynamics of almost automorphisms. We will make use of a description of the dynamics of individual elements of $\mathrm{AAut}(\mathcal{T})$, which is one of the subjects of [GL21]. Again, the proofs are given for $\mathcal{T} = \mathcal{T}_{d,k}$ but a careful reading shows that all arguments hold in the general case.

We need some definitions, following [GL21]: a *tree pair* is a tuple $(\kappa, \mathcal{T}_1, \mathcal{T}_2)$ where $\mathcal{T}_1, \mathcal{T}_2 \subseteq \mathcal{T}$ are finite complete trees with root v_0 and $\kappa\colon \mathcal{L}\mathcal{T}_1 \to \mathcal{L}\mathcal{T}_2$ is a bijection between the leaves of $\mathcal{T}_1$ and the leaves of $\mathcal{T}_2$. We say that a tree pair $(\kappa, \mathcal{T}_1, \mathcal{T}_2)$ is *associated* to an element $g \in \mathrm{AAut}(\mathcal{T})$ if g arises from an isomorphism of rooted forests $\overline{g}\colon \mathcal{T} \setminus \mathcal{T}_1 \to \mathcal{T} \setminus \mathcal{T}_2$ such that $\kappa = \overline{g}\big|_{\mathcal{L}\mathcal{T}_1}$. In this case $\{\partial\mathcal{T}_v\}_{v \in \mathcal{L}\mathcal{T}_1}$ is an admissible partition for g. We will only consider tree pairs that are associated to some almost automorphism of $\mathcal{T}$. All tree pairs are associated to almost automorphisms when $\mathcal{T} = \mathcal{T}_{d,k}$ but this is not true in general since the connected components of $\mathcal{T} \setminus \mathcal{T}_1$ and $\mathcal{T} \setminus \mathcal{T}_2$ need not be isomorphic.

Consider an orbit $\mathcal{O} = \{u_0, \dots, u_n\} \subseteq \mathcal{L}\mathcal{T}_1 \cup \mathcal{L}\mathcal{T}_2$ of the partial action of κ on $\mathcal{L}\mathcal{T}_1 \cup \mathcal{L}\mathcal{T}_2$, that is, $\mathcal{O}$ is such that $u_0, \dots, u_{n-1} \in \mathcal{L}\mathcal{T}_1$, $u_1, \dots, u_n \in \mathcal{L}\mathcal{T}_2$, $\kappa(u_j) = u_{j+1}$ for all $j = 0, \dots, n-1$ and either

- $u_0 \notin \mathcal{L}\mathcal{T}_2$ and $u_n \notin \mathcal{L}\mathcal{T}_1$, or
- $u_0 \in \mathcal{L}\mathcal{T}_2$, $u_n \in \mathcal{L}\mathcal{T}_1$ and $\kappa(u_n) = u_0$.

The orbit $\mathcal{O}$ is said to be

- an *attracting chain* if u_n is a descendant of u_0 in $\mathcal{T}$, in which case u_n is called the *attractor* of the orbit,
- a *repelling chain* if u_0 is a descendant of u_n in $\mathcal{T}$, in which case u_0 is called the *repeller* of the orbit,
- a *periodic chain* if $\kappa(u_n) = u_0$, and
- a *wandering chain* if $u_0 \notin \mathcal{T}_2$ (that is, u_0 is a descendant of some leaf in $\mathcal{L}\mathcal{T}_2$) and $u_n \notin \mathcal{T}_1$ (that is, u_n is a descendant of some leaf in $\mathcal{L}\mathcal{T}_1$).

These options are mutually exclusive. We say that $(\kappa, \mathcal{T}_1, \mathcal{T}_2)$ is a *revealing tree pair* if each connected component of $\mathcal{T}_1 \setminus \mathcal{T}_2$ contains a repeller and each connected component of $\mathcal{T}_2 \setminus \mathcal{T}_1$ contains an attractor. In this case, if $C \subseteq \mathcal{T}_1 \setminus \mathcal{T}_2$ is a connected component containing a repeller u_0, the orbit $\{u_0, \dots, u_n\}$ is such that u_n is the root of C and u_0 is the unique repeller in C. Similarly, if $C \subseteq \mathcal{T}_2 \setminus \mathcal{T}_1$ is a connected component containing an attractor u_n, the orbit $\{u_0, \dots, u_n\}$ is such that u_0 is the root of C and u_n is the unique attractor in C. Revealing tree pairs were introduced by O. Salazar-Díaz in [SD10] to describe the dynamics of individual elements of Thompson's V.

If $(\kappa, \mathcal{T}_1, \mathcal{T}_2)$ is a revealing tree pair, then every orbit $\mathcal{O}$ is an attracting, repelling, periodic or wandering chain: assume that $\mathcal{O}$ is not periodic, so it must end in an element $u_n \in \mathcal{LT}_2 \setminus \mathcal{LT}_1$ and begin in an element $u_0 \in \mathcal{LT}_1 \setminus \mathcal{LT}_2$. If $u_0 \in \mathcal{T}_2$, then u_0 is the root of its component of $\mathcal{T}_2 \setminus \mathcal{T}_1$ and u_n must be the attractor in this component. If $u_n \in \mathcal{T}_1$, then u_n is the root of its component in $\mathcal{T}_1 \setminus \mathcal{T}_2$ and u_0 must be the repeller in this component. If neither happen, then $\mathcal{O}$ is wandering.

Theorem 2.3 ([GL21, Lemma 2.17])**.** *Every element of* $\mathrm{AAut}(\mathcal{T})$ *is associated to some revealing tree pair.*

As a consequence we deduce the following corollary, the proof of which uses ideas present in [GL21, Section 3.1] (compare [HM19, Lemma 5.5] for Higman-Thompson's group V).

Corollary 2.4. *If $g \in \mathrm{AAut}(\mathcal{T})$ there is a partition $\partial\mathcal{T} = U_g \sqcup V_g$ into g-invariant clopen subsets such that U_g is equal to the open subset of $\xi \in \partial\mathcal{T}$ that admit a neighborhood $U \subseteq \partial\mathcal{T}$ such that some positive power of $g\big|_U$ is an isometry onto U. Moreover, the following properties are verified.*

- *There is a positive power of $g\big|_{U_g}$ that is an isometry (for the visual metric of $\partial\mathcal{T}$).*
- *There are finitely many g-periodic points in V_g, which we denote $\mathrm{Per}_{\mathrm{hyp}}(g)$.*
- *There is a partition $\mathrm{Per}_{\mathrm{hyp}}(g) = \mathrm{Per}_{\mathrm{rep}}(g) \sqcup \mathrm{Per}_{\mathrm{att}}(g)$ into* repelling *and* attracting *periodic points, such that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ so that for all $k \geq N$ we have*

$$g^k(V_g \setminus \mathrm{Per}_{\mathrm{rep}}(g)^\varepsilon) \subseteq \mathrm{Per}_{\mathrm{att}}(g)^\varepsilon \quad \textit{and} \quad g^{-k}(V_g \setminus \mathrm{Per}_{\mathrm{att}}(g)^\varepsilon) \subseteq \mathrm{Per}_{\mathrm{rep}}(g)^\varepsilon. \tag{2.1}$$

Proof. Let $(\kappa, \mathcal{T}_1, \mathcal{T}_2)$ be a revealing pair associated to g. Write $\widetilde{U}_g = \bigsqcup \partial\mathcal{T}_u$ where the union is over all vertices $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$ that are in periodic chains, and set $\widetilde{V}_g = \partial\mathcal{T} \setminus \widetilde{U}_g$. The sets $\widetilde{U}_g$ and $\widetilde{V}_g$ are g-invariant and clopen, and if $\partial\mathcal{T}_u \subseteq \widetilde{U}_g$ where $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$, there exists $n \in \mathbb{N}$ such that $g^n(\partial\mathcal{T}_u) = \partial\mathcal{T}_u$ and $g^n\big|_{\partial\mathcal{T}_u}$ is an isometry. By taking appropriate powers of g we see that there is a $k \in \mathbb{N}$ such that $g^k\big|_{\widetilde{U}_g}$ is an isometry.

Now let $\partial\mathcal{T}_u \subseteq \widetilde{V}_g$ where $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$ is in an attracting chain $\{u_0, \ldots, u_n\}$. We have that $g^n(\partial\mathcal{T}_{u_0}) \subsetneq \partial\mathcal{T}_{u_0}$ and $g^n\big|_{\partial\mathcal{T}_{u_0}}$ is a contraction, so in particular $g^n(\partial\mathcal{T}_u) \subsetneq \partial\mathcal{T}_u$ and $g^n\big|_{\partial\mathcal{T}_u}$ is a contraction. Thus there exists a unique g^n-fixed point $\xi_u \in \partial\mathcal{T}_u$, and which must also verify $\bigcap_{k\geq 0} g^{kn}(\partial\mathcal{T}_u) = \{\xi_u\}$. Set

$$\mathrm{Per}_{\mathrm{att}}(g) = \{\xi_u : u \text{ is in an attracting chain}\}.$$

If $\partial\mathcal{T}_u \subseteq \widetilde{V}_g$ where $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$ is in a repelling chain instead, the argument of the previous paragraph applied to g^{-1} shows that for some $n \in \mathbb{N}$ there exists a unique g^n-fixed point $\xi_u \in \partial\mathcal{T}_u$ such that $\bigcap_{k\geq 0} g^{-kn}(\partial\mathcal{T}_u) = \{\xi_u\}$. Set

$$\mathrm{Per}_{\mathrm{rep}}(g) = \{\xi_u : u \text{ is in a repelling chain}\}.$$

The sets $\mathrm{Per}_{\mathrm{att}}(g)$ and $\mathrm{Per}_{\mathrm{rep}}(g)$ are disjoint and finite. Moreover, their union gives all g-periodic points in $\widetilde{V}_g$ since any $\xi \in \partial\mathcal{T}_u$ where $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$ is in a wandering chain cannot be a g-periodic

point: indeed, if the orbit $\{u_0, \dots, u_n\}$ is wandering, then the connected component of $u_n \in \mathcal{LT}_2 \setminus \mathcal{T}_1$ in $\mathcal{T}_2 \setminus \mathcal{T}_1$ has a root $u_{\mathrm{a}} \in \mathcal{LT}_1$ which must be the first element of an attracting orbit. Thus

$$d\left(g^{n-k+j}\left(\partial \mathcal{T}_{u_k}\right), g^j\left(\xi_{u_{\mathrm{a}}}\right)\right) \xrightarrow{j\to\infty} 0$$

for every $k = 0, \dots, n$. In the same way, the root $u_{\mathrm{r}} \in \mathcal{LT}_2$ of the connected component of $u_0 \in \mathcal{LT}_1 \setminus \mathcal{T}_2$ in $\mathcal{T}_1 \setminus \mathcal{T}_2$ is the last element of a repelling orbit, so

$$d\left(g^{-k-j}\left(\partial \mathcal{T}_{u_k}\right), g^{-j}\left(\xi_{u_{\mathrm{r}}}\right)\right) \xrightarrow{j\to\infty} 0$$

for every $k = 0, \dots, n$. Since $\xi_{u_{\mathrm{r}}} \neq \xi_{u_{\mathrm{a}}}$, there are no periodic points in any $\partial \mathcal{T}_{u_k}$ for $k = 0, \dots, n$.

To prove that $\widetilde{V}_g$, $\mathrm{Per}_{\mathrm{att}}(g)$, $\mathrm{Per}_{\mathrm{rep}}(g)$ verify the last item of the corollary, by symmetry it suffices to prove the first statement in (2.1). To do this, take a clopen $U \subsetneq \widetilde{V}_g \setminus \mathrm{Per}_{\mathrm{rep}}(g)$ and $\varepsilon > 0$, and let $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$. If u is in an attracting or wandering chain, it is clear that there is an $N \in \mathbb{N}$ such that $g^k(U \cap \partial \mathcal{T}_u) \subseteq \mathrm{Per}_{\mathrm{att}}(g)^\varepsilon$ for all $k \geq N$. If u is in a repelling orbit $\{u_0, \dots, u_n\}$, then the equality

$$\bigcap_{k\geq 0} g^{-kn}(\partial \mathcal{T}_u) \cap (U \cap \partial \mathcal{T}_u) = \varnothing$$

shows that there is a $l \in \mathbb{N}$ such that $g^{-kn}(\partial \mathcal{T}_u) \cap (U \cap \partial \mathcal{T}_u) = \varnothing$ for all $k \geq l$. The sets $g^r(\partial \mathcal{T}_u)$, $r = 0, \dots, n-1$ are pairwise disjoint, and we conclude that $\partial \mathcal{T}_u \cap g^k(U \cap \partial \mathcal{T}_u) = \varnothing$ for all $k \geq nl$. Upon replacing U by $g^r(U)$ for some sufficiently big $r \in \mathbb{N}$ we may assume that $U \cap \partial \mathcal{T}_u$ is empty for all $u \in \mathcal{LT}_1 \cup \mathcal{LT}_2$ in a repelling orbit. Hence (2.1) holds for a sufficiently large $N \in \mathbb{N}$.

Up to now, $\widetilde{U}_g$, $\widetilde{V}_g$ satisfy all the required properties except that $\widetilde{V}_g$ could still contain some points that admit a neighborhood $U \subseteq \partial \mathcal{T}$ such that some positive power of $g\big|_U$ is an isometry onto U. Call the set of such points $\mathcal{I} \subseteq \widetilde{V}_g$. All elements in $\widetilde{V}_g \setminus (\mathrm{Per}_{\mathrm{rep}}(g) \sqcup \mathrm{Per}_{\mathrm{att}}(g))$ admit a neighborhood U such that $g^j(U) \cap U = \varnothing$ for all $j \neq 0$, so $\mathcal{I} \subseteq \mathrm{Per}_{\mathrm{rep}}(g) \sqcup \mathrm{Per}_{\mathrm{att}}(g)$ is finite. Thus setting $U_g = \widetilde{U}_g \sqcup \mathcal{I}$, $V_g = \widetilde{V}_g \setminus \mathcal{I}$ ensures that all properties in the statement of the corollary are verified. □

In contrast with the trees $\mathcal{T}_{d,k}$, the boundary of an arbitrary locally finite tree $\mathcal{T}$ may contain isolated points and the three items in the previous corollary do not specify U_g, V_g uniquely. The definition of U_g in the statement of the corollary gives a canonical definition in any case.

Vietoris topology. Given a topological space X, we denote by 2^X the space of closed subsets of X equipped with the *Vietoris topology*, defined as the topology with subbasis

$$\{K \in 2^X : K \cap U \neq \varnothing,\, K \cap V = \varnothing\}$$

where U, V range over all open subsets of X. If X is compact metrizable then 2^X is also compact metrizable [Eng89, Section 4.5], and in this case any action by homeomorphisms of a countable group on X induces naturally an action by homeomorphisms on 2^X [Eng89, Section 3.12].

3. Proofs

We first gather some useful lemmas.

Lemma 3.1 (Neumann's lemma, [Neu54, Lemma 4.1])**.** *Let G be a group acting on a set X and assume that the action has no finite orbits. Then for every pair of finite subsets A, B of X there exists an element $g \in G$ such that $g(A) \cap B$ is empty.*

Lemma 3.2 ([MM23, Proposition 1.16])**.** *Let G be a group acting by homeomorphisms on a compact metric space (X, d). Assume that*

i. there exists a positive integer p such that for any $\varepsilon > 0$ there exist non-empty finite sets $A, B \subseteq X$ of cardinality at most p with $g(X \setminus A^\varepsilon) \subseteq B^\varepsilon$ for some $g \in G$, and

ii. there are no finite G-orbits.

Then the action of G on X has a ping-pong pair.

Proof. Notice that condition (i) is equivalent to the existence of finite sets A, B of X that work for *any* $\varepsilon > 0$: a pair (A, B) of non-empty subsets of X of cardinality at most p is called a *contraction pair* if for every neighborhood U, V of A, B respectively there exists $g \in G$ such that $g(X \setminus U) \subseteq V$. Then condition (i) and the compactness of X imply that there exists a contraction pair.

Moreover, if (A, B) is a contraction pair and $u, v \in G$ then $(u(A), v(B))$ is a contraction pair too. Thus, by (ii) and Lemma 3.1, our contraction pair (A, B) can be taken such that A and B are disjoint.

Using (ii) and Lemma 3.1 again we deduce that there exist two contraction pairs (A_1, B_1), (A_2, B_2) where the A_1, A_2, B_1, B_2 are pairwise disjoint. If $U_i, V_i, i = 1, 2$ are pairwise disjoint neighborhoods of $A_i, B_i, i = 1, 2$ respectively, we can find $g_1, g_2 \in G$ such that $g_i(X \setminus U_i) \subseteq V_i$ for $i = 1, 2$. These constitute a ping-pong pair. □

Lemma 3.3. *Let G be a compact topological group and $g \in G$. For every neighborhood U of the identity there exists a strictly increasing sequence $(n_j)_{j \geq 0} \subseteq \mathbb{N}$ such that $g^{n_j} \in U$ for all $j \in \mathbb{N}$.*

Proof. Let μ be the normalized Haar measure on G, so that μ is a probability measure of complete support on G and left multiplication by g preserves μ. Let $V \subseteq U$ be an open neighborhood of the identity such that $V \cdot V^{-1} \subseteq U$. Since $\mu(V) > 0$, Poincaré's recurrence theorem implies that there is a strictly increasing sequence $(n_j)_{j \geq 0} \subseteq \mathbb{N}$ such that $\mu(V \cap g^{n_j} V) > 0$ for all $j \in \mathbb{N}$. In particular $g^{n_j} \in V \cdot V^{-1} \subseteq U$ for all $j \in \mathbb{N}$. □

The core of the proof of Theorem A is the following statement, which uses ideas from [LBMB22, Proposition 5.2] and implies the extreme contraction properties required by Lemma 3.2. For an almost automorphism $g \in \mathrm{AAut}(\mathcal{T})$ we use the same notation as Corollary 2.4, so we write $U_g \subseteq \partial\mathcal{T}$ for the stable set of g and $\mathrm{Per}_{\mathrm{hyp}}(g)$, $\mathrm{Per}_{\mathrm{rep}}(g)$ for the hyperbolic and repelling points of g respectively.

Recall that we denote by $2^{\partial\mathcal{T}}$ the compact metrizable space of all closed subsets of $\partial\mathcal{T}$ equipped with the Vietoris topology.

Proposition 3.4. *Let H be a subgroup of $\mathrm{AAut}(\mathcal{T})$ and assume that $\bigcap_{h\in H} U_h$ is empty. Then there exists a finite set $B \subseteq \partial\mathcal{T}$ such that for every $\varepsilon > 0$ there is a $h \in H$ with $h(\partial\mathcal{T} \setminus B^\varepsilon) \subseteq B^\varepsilon$.*

Proof. By compactness there exist elements $h_1, \dots, h_k \in H$ such that $\bigcap_{j=1}^k U_{h_j}$ is empty. By taking powers of the h_j we can assume that every $h_j\big|_{U_j}$ is an isometry. Set $B = \bigcup_{j=1}^k \mathrm{Per}_{\mathrm{hyp}}(h_j)$ and $C_1 = \partial\mathcal{T} \setminus B^\varepsilon$.

Now h_1 restricted to U_{h_1} is an isometry. Since isometry groups of compact spaces are compact for the compact-open topology, Lemma 3.3 and a diagonal argument shows that there exists a strictly increasing sequence $(n_j)_{j\geq 0} \subseteq \mathbb{N}$ such that

$$\sup_{x\in U_{h_1}} d\left(h_1^{n_j}(x), x\right) \leq \frac{1}{j}$$

for all $j \in \mathbb{N}$.

For any closed $D \subseteq \partial\mathcal{T}$ denote by $\overline{\mathrm{Orb}_H(D)}$ the closure of the H-orbit of D in $2^{\partial\mathcal{T}}$. By taking a limit point of $(h_1^{n_j}(C_1))_{j\geq 0}$ in $2^{\partial\mathcal{T}}$ we obtain a closed $C_2 \in \overline{\mathrm{Orb}_H(C_1)}$ with

$$C_2 \subseteq (C_1 \cap U_{h_1}) \sqcup \mathrm{Per}_{\mathrm{hyp}}(h_1).$$

Lemma 3.3 applied to h_2 gives again the existence of a sequence $(n'_j)_{j\geq 0} \subseteq \mathbb{N}$ such that

$$\sup_{x\in U_{h_2}} d\left(h_2^{n'_j}(x), x\right) \leq \frac{1}{j}$$

for all $j \in \mathbb{N}$. The finite set $\mathrm{Per}_{\mathrm{rep}}(h_2)$ can only intersect C_2 in $\mathrm{Per}_{\mathrm{hyp}}(h_1)$, so by taking again a limit point of $(h_2^{n'_j}(C_2))_{j\geq 0}$ we obtain a closed $C_3 \in \overline{\mathrm{Orb}_H(C_2)} \subseteq \overline{\mathrm{Orb}_H(C_1)}$ with

$$C_3 \subseteq (C_2 \cap U_{h_2}) \sqcup \mathrm{Per}_{\mathrm{hyp}}(h_2) \subseteq (C_1 \cap U_{h_1} \cap U_{h_2}) \sqcup (\mathrm{Per}_{\mathrm{hyp}}(h_1) \cup \mathrm{Per}_{\mathrm{hyp}}(h_2)).$$

Notice that, again, the finite set $\mathrm{Per}_{\mathrm{rep}}(h_3)$ can only intersect C_3 in $\mathrm{Per}_{\mathrm{hyp}}(h_1) \cup \mathrm{Per}_{\mathrm{hyp}}(h_2)$.

By iterating this argument inductively we produce, for every $j = 2, \dots, k+1$, a closed subset $C_j \in \overline{\mathrm{Orb}_H(C_{j-1})} \subseteq \overline{\mathrm{Orb}_H(C_1)}$ such that

$$C_j \subseteq (C_{j-1} \cap U_{h_{j-1}}) \sqcup \mathrm{Per}_{\mathrm{hyp}}(h_{j-1}) \subseteq (C_1 \cap U_{h_1} \cap \dots \cap U_{h_{j-1}}) \sqcup \left(\bigcup_{i=1}^{j-1} \mathrm{Per}_{\mathrm{hyp}}(h_i)\right).$$

In particular $C_{k+1} \subseteq \bigcup_{i=1}^k \mathrm{Per}_{\mathrm{hyp}}(h_i) = B$, and we are done. □

Proof of Theorem A. Take a subgroup H of $\mathrm{AAut}(\mathcal{T})$ and suppose that H does not preserve a probability measure on $\partial\mathcal{T}$. If every finitely generated subgroup of H preserves a probability measure on $\partial\mathcal{T}$, the compactness of $\mathrm{Prob}(\partial\mathcal{T})$ equipped with the weak-$*$ topology implies that H preserves a probability measure on $\partial\mathcal{T}$. We may assume then that H is finitely generated.

Suppose that the set $U_H = \bigcap_{h\in H} U_h$ is non-empty. Since every U_h is dynamically defined (see the definition of U_h in Corollary 2.4), we have that U_H is H-invariant. The closed set U_H may be written as $\partial\mathcal{T}'$ for some rooted subtree $\mathcal{T}' \subseteq \mathcal{T}$, and we equip U_H with the restriction of the visual metric of $\partial\mathcal{T}$, which coincides with the visual metric of $\partial\mathcal{T}'$. Every element of H has a proper power

that acts as an isometry on $\partial\mathcal{T}'$, so the action of H on $\partial\mathcal{T}'$ is by elliptic almost automorphisms of $\mathcal{T}'$. Since H is finitely generated, Proposition 2.1 shows that the image of H in $\mathrm{AAut}(\mathcal{T}')$ is relatively compact. Thus H preserves a measure on $\partial\mathcal{T}' \subseteq \partial\mathcal{T}$, a contradiction. We conclude that U_H is empty, and in this case Proposition 3.4 and Lemma 3.2 imply that there exists a ping-pong pair for the action of H. □

Remark. The proof of Theorem A shows a slightly stronger statement, namely that for any finitely generated subgroup H of $\mathrm{AAut}(\mathcal{T})$, either H contains a ping-pong pair or there exists a non-empty closed set $C \subseteq \partial\mathcal{T}$ such that the action of H on C is equicontinuous.

Recall that $V_{\mathcal{T}}$ denotes the Higman-Thompson group associated to $\mathcal{T}$.

Proof of Theorem B. Fix a linear order on $\partial\mathcal{T}$ as in the definition of $V_{\mathcal{T}}$. Take H a finitely generated subgroup of $V_{\mathcal{T}}$ and suppose that H acts without finite orbits on $\partial\mathcal{T}$. If $U_H = \bigcap_{h\in H} U_h$ is non-empty, consider again a rooted subtree $\mathcal{T}' \subseteq \mathcal{T}$ such that $U_H = \partial\mathcal{T}'$, so H acts on U_H by elliptic Higman-Thompson elements of $\mathrm{AAut}(\mathcal{T}')$ for the induced order on $\partial\mathcal{T}'$. Corollary 2.2 shows that the image of H in $\mathrm{AAut}(\mathcal{T}')$ is finite, and thus there exists a finite H-orbit in U_H. This is a contradiction, hence U_H is empty. Again Proposition 3.4 and Lemma 3.2 show that there exists a ping-pong pair for the action of H. □

4. Open questions

We finish with some hopefully tractable open questions.

Question. The whole homeomorphism group of the Cantor space does not verify the dynamical Tits alternative.[1] Nevertheless, Grigorchuk-Nekrashevych-Sushchansky introduce in [GNS00] the group $\mathcal{R}$ of homeomorphisms of Cantor space defined by asynchronous transducers, which is known [BB17] to contain the higher-dimensional Brin-Thompson groups from [Bri04]. On the other hand, the automorphism group of a Higman-Thompson group $V_{d,k}$ coincides with the subgroup of bi-synchronizing transducers inside $\mathcal{R}$ [BCM$^+$24]. Does the natural action of any of these groups on the Cantor set satisfy the dynamical Tits alternative?

Question. Let G be a group of homeomorphisms of a compact topological space X such that its groupoid of germs is hyperbolic in the sense of V. Nekrashevych, see [Nek15]. Does the action of G on X satisfy the dynamical Tits alternative? Theorem A shows that this is true for the groups V_G, whose groupoid of germs is hyperbolic whenever G is a contracting self-similar group.

[1] Take any non-amenable countable group G without non-abelian free subgroups. Since G is non-amenable, there is a subshift $X \subseteq \{0,1\}^G$ with no invariant probability measures [FKSV25, Theorem 5.13]. X cannot have isolated points and is thus homeomorphic to a Cantor set, and the action $G \curvearrowright X$ does not satisfy the alternative.

References

[AS22] Richard Aoun and Cagri Sert, *Random walks on hyperbolic spaces: concentration inequalities and probabilistic Tits alternative*, Probability Theory and Related Fields **184** (2022), no. 1-2, pp. 323–365. (Cited on page 2).

[BB17] James Belk and Collin Bleak, *Some undecidability results for asynchronous transducers and the Brin-Thompson group* $2V$, Transactions of the American Mathematical Society **369** (2017), no. 5, pp. 3157–3172. (Cited on page 10).

[BCGM12] Uri Bader, Pierre-Emmanuel Caprace, Tsachik Gelander, and Shahar Mozes, *Simple groups without lattices*, Bulletin of the London Mathematical Society **44** (2012), no. 1, pp. 55–67. (Cited on page 3).

[BCM+24] Collin Bleak, Peter Cameron, Yonah Maissel, Andrés Navas, and Feyishayo Olukoya, *The further chameleon groups of Richard Thompson and Graham Higman: automorphisms via dynamics for the Higman-Thompson groups* $G_{n,r}$, Memoirs of the American Mathematical Society **301** (2024), no. 1510, pp. viii+94. (Cited on page 10).

[Bri04] Matthew G. Brin, *Higher dimensional Thompson groups*, Geometriae Dedicata **108** (2004), pp. 163–192. (Cited on page 10).

[CCMT15] Pierre-Emmanuel Caprace, Yves Cornulier, Nicolas Monod, and Romain Tessera, *Amenable hyperbolic groups*, Journal of the European Mathematical Society **17** (2015), no. 11, pp. 2903–2947. (Cited on page 2).

[CFP96] James W. Cannon, William J. Floyd, and Walter R. Parry, *Introductory notes on Richard Thompson's groups*, L'Enseignement Mathématique. 2e Série **42** (1996), no. 3-4, pp. 215–256. (Cited on page 1).

[dlH00] Pierre de la Harpe, *Topics in geometric group theory*, Chicago Lectures in Mathematics, University of Chicago Press, Chicago, IL, 2000. (Cited on page 1).

[DM18] Bruno Duchesne and Nicolas Monod, *Group actions on dendrites and curves*, Annales de l'Institut Fourier **68** (2018), no. 5, pp. 2277–2309. (Cited on page 2).

[Eng89] Ryszard Engelking, *General topology*, second ed., Sigma Series in Pure Mathematics, vol. 6, Heldermann Verlag, Berlin, 1989, Translated from the Polish by the author. (Cited on page 7).

[FKSV25] Joshua Frisch, Alexander Kechris, Forte Shinko, and Zoltán Vidnyánszky, *Realizations of countable Borel equivalence relations* (2025 preprint), arXiv:2109.12486, to appear in Memoirs of the American Mathematical Society. (Cited on page 10).

[GL18] Lukasz Garncarek and Nir Lazarovich, *The Neretin groups*, New directions in locally compact groups, London Mathematical Society Lecture Note Series, vol. 447, Cambridge University Press, Cambridge, 2018, pp. 131–144. (Cited on page 3).

[GL21] Gil Goffer and Waltraud Lederle, *Conjugacy and dynamics in almost automorphism groups of trees*, International Journal of Algebra and Computation **31** (2021), no. 8, pp. 1497–1545. (Cited on pages 3, 5, 6).

[GNS00] Rostislav I. Grigorchuk, Volodymyr V. Nekrashevich, and Vitaly I. Sushchanskiĭ, *Automata, dynamical systems, and groups*, Trudy Matematicheskogo Instituta Imeni V. A. Steklova **231** (2000), no. Din. Sist., Avtom. i Beskon. Gruppy, pp. 134–214. (Cited on page 10).

[Gri80] Rostislav I. Grigorchuk, *On Burnside's problem on periodic groups*, Akademiya Nauk SSSR. Funktsional nyĭ Analiz i ego Prilozheniya **14** (1980), no. 1, pp. 53–54. (Cited on page 1).

[HM19] Sebastián Hurtado and Emmanuel Militon, *Distortion and Tits alternative in smooth mapping class groups*, Transactions of the American Mathematical Society **371** (2019), no. 12, pp. 8587–8623. (Cited on pages 2, 3, 6).

[Kap99] Christophe Kapoudjian, *Simplicity of Neretin's group of spheromorphisms*, Annales de l'Institut Fourier **49** (1999), no. 4, pp. 1225–1240. (Cited on page 3).

[LB17] Adrien Le Boudec, *Compact presentability of tree almost automorphism groups*, Annales de l'Institut Fourier **67** (2017), no. 1, pp. 329–365. (Cited on page 3).

[LBMB22] Adrien Le Boudec and Nicolás Matte Bon, *Confined subgroups and high transitivity*, Annales Henri Lebesgue **5** (2022), pp. 491–522. (Cited on page 8).

[LBW19] Adrien Le Boudec and Phillip Wesolek, *Commensurated subgroups in tree almost automorphism groups*, Groups, Geometry, and Dynamics **13** (2019), no. 1, pp. 1–30. (Cited on pages 3, 4, 5).

[Led20] Waltraud Lederle, *Topological full groups and t.d.l.c. completions of Thompson's V*, Mathematische Annalen **378** (2020), no. 3-4, pp. 1415–1434. (Cited on page 3).

[Mar00] Gregory A. Margulis, *Free subgroups of the homeomorphism group of the circle*, Comptes Rendus de l'Académie des Sciences. Série I. Mathématique **331** (2000), no. 9, pp. 669–674. (Cited on page 2).

[MM23] Dominique Malicet and Emmanuel Militon, *Random actions of homeomorphisms of Cantor sets embedded in a line and Tits alternative* (2023 preprint), arXiv:2304.08070, to appear in Annales Henri Lebesgue. (Cited on pages 2, 8).

[Nek15] Volodymyr V. Nekrashevych, *Hyperbolic groupoids and duality*, Memoirs of the American Mathematical Society **237** (2015), no. 1122, pp. v+105. (Cited on page 10).

[Nek18] ———, *Finitely presented groups associated with expanding maps*, Geometric and cohomological group theory, London Mathematical Society Lecture Note Series, vol. 444, Cambridge University Press, Cambridge, 2018, pp. 115–171. (Cited on page 3).

[Neu54] Bernhard H. Neumann, *Groups covered by permutable subsets*, The Journal of the London Mathematical Society **29** (1954), pp. 236–248. (Cited on page 8).

[SD10] Olga Patricia Salazar-Díaz, *Thompson's group V from a dynamical viewpoint*, International Journal of Algebra and Computation **20** (2010), no. 1, pp. 39–70. (Cited on pages 3, 5).

[Tit70] Jacques Tits, *Sur le groupe des automorphismes d'un arbre*, Essays on Topology and Related Topics (Mémoires dédiés à Georges de Rham), Springer, New York-Berlin, 1970, pp. 188–211. (Cited on page 2).

[Tit72] ———, *Free subgroups in linear groups*, Journal of Algebra **20** (1972), pp. 250–270. (Cited on page 1).

[Zhe19] Tianyi Zheng, *Neretin groups admit no non-trivial invariant random subgroups* (2019 preprint), arXiv:1905.07605. (Cited on page 3).

Institut Camille Jordan, Université Claude Bernard Lyon 1, 43 boulevard du 11 novembre 1918, 69622 Villeurbanne, France.

Email address: gilabert@math.univ-lyon1.fr // martingilabertvio@gmail.com